UNIT 4

Tools for Web Attack and Security Testers

Network Scanning Tools

- Used to identify live hosts, open ports, and services
- Helps attackers/testers understand target system

Examples:

- Nmap – Port scanning, OS detection
- Netcat – Banner grabbing, network debugging

Vulnerability Scanners

- Detect weaknesses in web applications and systems
- Helps in finding security flaws automatically

Examples:

- Nessus – Finds vulnerabilities and misconfigurations
- OpenVAS – Open-source vulnerability scanner

Web Application Testing Tools

- Used to test web apps for security issues like SQL injection, XSS

Examples:

- Burp Suite – Intercept requests, modify traffic
- OWASP ZAP – Automated vulnerability detection

Password Cracking Tools

- Used to break passwords using different techniques

Examples:

- John the Ripper – Dictionary and brute force attack
- Hydra – Online password cracking tool

Packet Sniffing Tools

- Capture and analyze network traffic
- Used to detect sensitive data transfer

Examples:

- Wireshark – Detailed packet analysis
- Tcpdump – Command-line packet capture

Exploitation Tools

- Used to exploit vulnerabilities in systems

Examples:

- Metasploit Framework – Develop and execute exploits

Social Engineering Tools

- Used to trick users into revealing information

Examples:

- SET (Social Engineering Toolkit) – Phishing attacks

Wireless Testing Tools

- Used for testing Wi-Fi network security

Examples:

- Aircrack-ng – Crack Wi-Fi passwords
- Kismet – Wireless network detection

Hacking Wireless Networks/ Wireless Hacking

Definition

- Wireless hacking means accessing a Wi-Fi network without permission
- Done by attackers or security testers to find weaknesses

Types of Wireless Networks

- **Open Network** – No password, easy to hack
- **WEP (Wired Equivalent Privacy)** – Very weak security
- **WPA / WPA2** – Stronger security
- **WPA3** – Latest and most secure

Basic Concepts

- **SSID** – Name of Wi-Fi network
- **BSSID** – MAC address of router
- **Channel** – Frequency used for communication
- **Handshake** – Authentication process between device and router

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Wireless Hacking Techniques

a) Packet Sniffing

- Capturing data packets from network
- Used to analyze traffic

b) Brute Force Attack

- Trying all possible password combinations

c) Dictionary Attack

- Uses list of common passwords

d) Deauthentication Attack

- Disconnects users from network
- Forces reconnection to capture handshake

e) Evil Twin Attack

- Fake Wi-Fi network created
- Users connect and attacker steals data

f) WPS Attack

- Targets WPS feature of router
- Easily cracks PIN-based security

Tools Used in Wireless Hacking

Aircrack-ng

- Used to crack WEP/WPA passwords

Wireshark

- Used for packet analysis

Kismet

- Detects wireless networks

Reaver

- Attacks WPS-enabled routers

Fern WiFi Cracker

- GUI-based wireless hacking tool

Steps in Wireless Hacking

1. Identify available Wi-Fi networks
2. Select target network
3. Capture packets (handshake)
4. Perform attack (brute force/dictionary)
5. Crack password
6. Gain access

Risks of Wireless Hacking

- Unauthorized access
- Data theft
- Identity theft
- Network misuse

Prevention Methods

- Use **WPA3 encryption**
- Set strong passwords
- Disable WPS
- Hide SSID
- Use firewall and VPN
- Regularly update router firmware

UNIT 5

Intrusion Detection and Prevention System (IDPS)

- IDPS is a security system that monitors network and system activities
- It detects suspicious behavior and prevents cyber attacks
- Used to protect confidentiality, integrity, and availability of data

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Need for IDPS

- Firewalls cannot detect all attacks
- Increasing number of cyber threats
- Need for real-time monitoring
- Helps in early detection and response

Types of IDPS

a) Network-based IDPS (NIDPS)

- Monitors traffic across the entire network
- Placed at gateways or routers
- Detects attacks like DoS, port scanning

b) Host-based IDPS (HIDPS)

- Installed on individual systems
- Monitors system logs, file changes
- Detects insider attacks

c) Wireless IDPS (WIDPS)

- Monitors wireless networks
- Detects unauthorized access points (rogue APs)
- Protects Wi-Fi networks

d) Hybrid IDPS

- Combines NIDPS and HIDPS
- Provides complete security coverage

Detection Techniques

a) Signature-Based Detection

- Uses database of known attack patterns (signatures)
- Fast and accurate for known attacks
- Cannot detect new (zero-day) attacks

b) Anomaly-Based Detection

- Compares normal behavior with current activity
- Detects unusual patterns
- Can detect new attacks
- May produce false alarms

c) Stateful Protocol Analysis

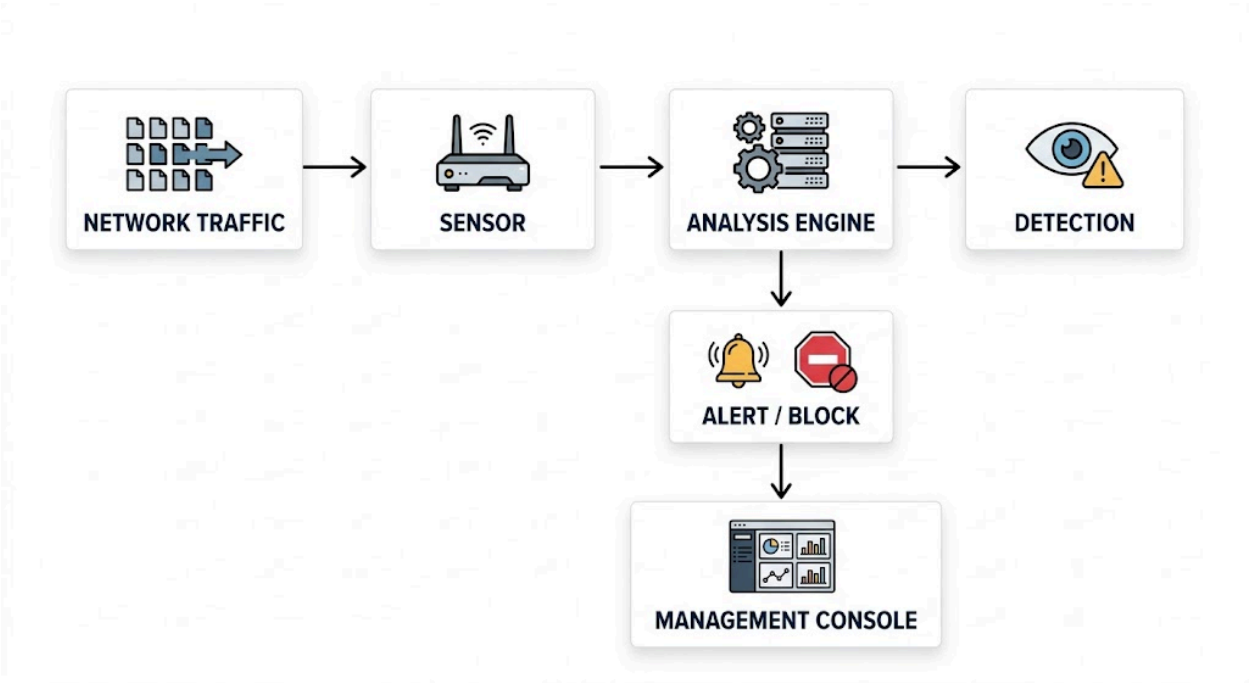
- Monitors protocol behavior
- Detects deviations from standard rules

Components of IDPS

- **Sensors/Agents**
  - Collect data from network or host
- **Analysis Engine**
  - Examines data to detect threats
- **Signature Database**
  - Stores known attack patterns
- **Management Console**
  - Used by admin to monitor and control
- **Response System**
  - Takes action like blocking or alerting

Working of IDPS (Step-by-Step Flow)

1. Data is captured from network or host
2. Data is analyzed using detection methods
3. Compared with signatures or behavior patterns
4. If attack detected → alert generated
5. IPS blocks attack automatically (if enabled)
6. Logs are stored for future analysis,



IDS vs IPS (Detailed Difference)

Feature	IDS	IPS
Full Form	Intrusion Detection System	Intrusion Prevention System
Function	Detects attacks	Detects and prevents attacks
Response	Alert only	Automatic action
Placement	Passive	Inline
Risk	No blocking	May block valid traffic

Deployment Modes

- **Inline Mode** – IPS actively blocks traffic
- **Passive Mode** – IDS monitors and alerts only

Advantages

- Real-time threat detection
- Prevents attacks automatically
- Improves network security
- Helps in forensic analysis
- Detects insider threats

Tools for Firewalls and Routers – 16 Marks

Firewall Tools

a) pfSense

- Free and open-source firewall
- Used to control and protect network
- Supports VPN and monitoring

b) Cisco ASA

- Hardware firewall device
- Gives strong security
- Used in big companies

c) iptables

- Firewall tool in Linux
- Works using commands
- Filters incoming and outgoing traffic

d) Windows Defender Firewall

- Built-in firewall in Windows
- Easy to use
- Provides basic protection

e) ZoneAlarm

- Used for personal computers
- Blocks unauthorized access

Router Tools

a) Cisco IOS

- Operating system for Cisco routers
- Helps in routing and security

b) OpenWrt

- Open-source router software
- Gives more control and security

c) DD-WRT

- Improves router performance
- Adds extra security features

d) MikroTik RouterOS

- Advanced router software
- Used for managing network traffic

Monitoring Tools (Used with Firewalls & Routers)

a) Wireshark

- Captures network data
- Helps to analyze traffic

b) Nagios

- Monitors network devices
- Sends alerts if problem occurs

c) SolarWinds Network Performance Monitor

- Shows network performance
- Helps manage routers and firewalls

Functions of Firewall

- Blocks unauthorized users
- Filters network traffic
- Protects system from attacks

Functions of Router

- Connects different networks
- Sends data from one network to another
- Controls data flow

Features

- Packet filtering
- Access control
- Traffic monitoring
- VPN support

Advantages

- Improves security
- Controls network traffic
- Protects from hackers

UNIT 3

Vulnerability Assessment

- Vulnerability assessment means finding security weaknesses in a system, network, or application
- Helps to identify problems before attackers use them

Purpose

- Find security gaps
- Reduce risk of attacks
- Improve system security
- Protect data and network

Types of Vulnerability Assessment

a) Network-based Assessment

- Checks network devices like routers and switches
- Finds open ports and weak configurations

b) Host-based Assessment

- Checks individual systems (computers/servers)
- Finds software vulnerabilities

c) Web Application Assessment

- Tests websites and web apps
- Detects issues like SQL injection, XSS

d) Wireless Assessment

- Checks Wi-Fi networks
- Finds weak passwords and unauthorized access

Steps in Vulnerability Assessment

1. **Planning**
    - Decide what system to test
2. **Scanning**
    - Use tools to scan for vulnerabilities
3. **Analysis**
    - Study the results
4. **Risk Assessment**
    - Find how serious each vulnerability is
5. **Reporting**
    - Prepare report with findings
6. **Remediation**
    - Fix the vulnerabilities

Tools Used

a) Nessus

- Popular scanning tool
- Finds many types of vulnerabilities

b) OpenVAS

- Free tool
- Detects security issues

c) Nikto

- Tests web servers
- Finds misconfigurations

d) Qualys

- Cloud-based scanning tool
- Provides detailed reports

Common Vulnerabilities

- Weak passwords
- Outdated software
- Open ports
- Misconfiguration
- Unpatched systems

Advantages

- Early detection of security issues
- Prevents attacks
- Improves system performance
- Helps in risk management

Difference between Vulnerability Assessment and Penetration Testing

Feature	Vulnerability Assessment	Penetration Testing
Purpose	Find weaknesses	Exploit weaknesses
Method	Scanning	Real attack simulation
Result	List of vulnerabilities	Shows real impact

Tools for Identifying Vulnerabilities in Windows & Linux OS

- Vulnerability identification tools are used to find security weaknesses in operating systems
- Helps to detect problems before attackers exploit them

Tools for Windows OS

a) Microsoft Baseline Security Analyzer

- Checks Windows system for missing updates
- Finds weak passwords and misconfigurations

b) Nessus

- Scans Windows systems for vulnerabilities
- Gives detailed reports

c) Qualys

- Cloud-based scanning tool
- Detects vulnerabilities in Windows systems

d) OpenVAS

- Free tool
- Scans Windows for security issues

e) Wireshark

- Captures network traffic
- Helps detect suspicious activity

Tools for Linux OS

a) Lynis

- Performs security audit on Linux systems
- Detects vulnerabilities and configuration issues

b) OpenVAS

- Works in Linux environment
- Finds system and network vulnerabilities

c) Nmap

- Scans open ports and services
- Helps identify weak points

d) Nikto

- Scans web servers on Linux
- Detects vulnerabilities

e) ClamAV

- Detects malware in Linux
- Helps in vulnerability protection

Common Vulnerabilities in OS

- Weak passwords
- Missing updates (patches)
- Open ports
- Misconfiguration
- Malware infections

Working Process

1. Select system (Windows/Linux)
2. Run scanning tool
3. Identify vulnerabilities
4. Analyze severity
5. Fix the issues

Advantages

- Early detection of problems
- Improves system security
- Reduces risk of attack